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Via Electronic Submission

Assistant Secretary for Technology Policy
Office of the National Coordinator for Health Information Technology
Attention: HHS Health Sector AI RFI, Mary E. Switzer Building
Mail Stop: 7033A, 330 C Street SW, Washington, DC 20201

RE: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

To whom it may concern,

On behalf of the Medical Device Manufacturers Association (MDMA), below please find comments on the Request for Information, *Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care*.

MDMA is a national trade association that provides educational and advocacy assistance to hundreds of innovative companies in the field of medical technology. Our members, the majority of which are small to mid-sized medical device companies, have a strong record of delivering breakthrough therapies to treat chronic diseases and life-threatening conditions while lowering the cost of care. MDMA's mission is to ensure that patients have timely access to the latest advancements of safe and effective medical technologies that improve health outcomes. We appreciate the opportunity to provide these comments, and we look forward to ongoing collaboration with the Assistant Secretary for Technology Policy and the Office of the National Coordinator for Health Information Technology on this important issue.

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

AI has been incorporated into regulated medical devices for three decades, and the existing Food and Drug Administration (FDA) medical device regulatory requirements remain appropriate for these devices. As more novel AI technologies emerge in medical devices (e.g., generative AI and agentic AI), it is important to assess whether any specific updates to the regulatory framework may be needed to address their unique characteristics.

We encourage FDA to maintain a risk-based, technology-agnostic approach that is consistent with its longstanding device framework. Regulatory oversight should be proportionate to the clinical risk posed by the device and the degree of AI autonomy or system influence, rather than triggered solely by the presence of AI. Devices incorporating limited AI functionality should not be subject to the same expectations as systems that generate outputs or modify

clinically actionable outputs. This approach will foster continued innovation while preserving patient safety.

Furthermore, it is essential that the FDA remains the primary regulatory authority for AI-enabled medical devices. Establishing separate regulatory requirements or oversight bodies for AI would add significant complexity and regulatory burden, hindering innovation without providing additional benefits to patient safety.

We also urge the Department of Health and Human Services (HHS) to advance standards and interoperability by prioritizing evaluation and adoption of AI-relevant standards, including model lifecycle, data/metadata quality, and transparency. This will help reduce fragmentation for both providers and manufacturers.

HHS should advance US leadership in internationally recognized consensus standards and regulatory cooperation mechanisms for AI-enabled health technologies to reduce duplicative requirements across markets, strengthen American competitiveness, and accelerate timely patient access, while preserving FDA's risk-based statutory authority.

Unclear regulatory boundaries exist between the Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology (ASTP/ONC), decision support interventions (DSIs), and predictive decision support interventions (pDSIs), administrative AI, and workflow optimization tools. There is also fragmented oversight between ASTP/ONC, FDA clinical decision support (CDS), and the Centers for Medicare and Medicaid Services (CMS), which creates uncertainty for developers seeking to understand applicable requirements across the AI lifecycle and highlights the need for better cross-agency coordination to harmonize expectations for AI driven health technologies.

Lack of predictable expectations for model updates, learning systems, and post-deployment monitoring is another barrier to private sector innovation and clinical adoption. Additionally, misalignment between ASTP/ONC lifecycle frameworks and medical device regulatory requirements result in unclear regulatory expectations. Specifically, it would provide clarity if the marketing requirements for FDA regulated devices were the *only* regulatory requirements, without any overlapping requirements from ONC.

We'd also encourage HHS to advance regulatory and reimbursement approaches for inpatient and hospital outpatient settings in addition to decentralized setting since in these settings, AI enabled medical technologies can have a significant impact on both cost and care delivery.

Finally, unpredictable and unclear reimbursement pathways present barriers to private sector innovation in AI for health care and its adoption and use in clinical care. Because the coverage, coding, and payment methodologies of CMS and other payers have not kept pace with technology, developers face significant risk and uncertainty with respect to whether physicians, hospitals, and other providers will be appropriately and adequately reimbursed for utilizing their products, which can slow and even stifle innovation.

2. ***What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.***

AI is increasingly embedded within regulated medical devices, diagnostics, and integrated hardware-software ecosystems to help support clinical decision making and therapeutic and diagnostic interventions. It can allow for improved visualization, patient-specific plans, real-time intraoperative decision support, and holds the potential to improve clinical outcomes while reducing workforce burdens across the care team, including surgeons and medical staff.

The challenge is that existing reimbursement systems are tied to the costs of earlier versions of supplies and equipment that do not reflect the value, complexity, or ongoing costs of AI-enabled systems. While billing codes may be used in some cases to report the procedures performed, the incremental value and costs of AI-powered solutions are not captured in payment rates. We urge HHS/CMS to consider reimbursement mechanisms that facilitate adoption of these new technologies.

A reliable and predictable pathway to payment for AI and algorithmic tools is critical. Lack of such a pathway creates uncertainty for manufacturers and investors alike, limiting the development and adoption of AI-driven solutions. Venture investors consider a clear coding and reimbursement strategy to be the single most important factor when deciding whether to fund AI-based medical device solutions. For established medical technology manufacturers, this uncertainty also constrains long-term capital planning, lifecycle software investment, and U.S.-based innovation. Without a reliable pathway to payment, these innovations risk reduced venture investment, slower adoption, and limited market uptake.

Further, we support the reimbursement for AI enablement via CPT or other codes that are tied to demonstrable improvements in quality of care, reduction of errors, efficiency gains, predictability, reproducibility, and productivity improvements resulting from AI use (consistent with ACLA and American Medical Association positions). These mechanisms should recognize AI functionality when it demonstrably contributes to improved outcomes, safety, or efficiency whether as an integral component of medical devices, or as stand-alone clinical tools. This might include supporting value-based reimbursement pathways, reimbursing as part of care bundles, risk-based contracts, quality improvement programs, and outcome-based reimbursement.

We recommend that CMS/ CMMI pilot an AI reimbursement model specifically for AI enabled diagnostics, robotics, laboratory applications, operational improvements, outcome metrics, and other measurable clinical and system-level benefits. Pilots should include hospital and ambulatory surgical center settings and should account for the unique needs of capital-intensive technologies that rely on continuous software improvement, with a clearly defined pathway to transition successful pilots into permanent coding, payment, or coverage mechanisms to accelerate sustainable reimbursement and broader adoption. CMS should ensure that any new AI-specific reimbursement models supplement existing pathways rather than supplant technologies that have already secured coverage. For AI-powered solutions with established coding and payment, these existing frameworks must be preserved to maintain market stability and patient access.

When new medical technologies emerge that deploy AI and have no predicate 510(k), it would be helpful to have a more predictable review process than the one currently in place for the De Novo pathway. The Medical Device User Fee Amendments (MDUFA) De Novo target is to have only 70% of submissions meet a 150-day review target. Further, to date,

only about 50% of De Novos have obtained a positive determination allowing marketing the subject product. The de novo process thus is an uncertain one that makes projecting and preparing for product launch challenging. This level of unpredictability is particularly difficult for AI-enabled platforms that require synchronized regulatory approval, reimbursement planning, and clinical adoption. We urge HHS/FDA to implement improvements to the De Novo process to address these issues, including issuance of clearer evidentiary expectations, greater transparency on review timelines, and more consistent application of standards.

While not exclusive but certainly related to AI product reviews, manufacturers need clearer guidance on the acceptability of data originating outside of the US. Moreover, further clarification on what demographic and site differentiations are required, both for training and testing, also would help define the type of data required for submissions and improve the predictability of submission outcomes.

The CMS HTI-1 rule has been interpreted publicly by CMS to both apply and not apply to products regulated as medical devices if it is training algorithms on data from Electronic Health Records. This interpretation extends to medical devices already reviewed and cleared/approved by the FDA via the 510(k) or PMA process, which is duplicative, burdensome and redundant. Modification of this rule should be prioritized to avoid needless duplicative and burdensome review for products already determined to be safe and effective by the FDA. Further clarification on the scope of the rule would ensure that medical device manufacturers understand all the requirements that apply to their AI-driven medical devices not subject to FDA review.

Clinical Laboratory Improvement Amendments (CLIA) AI requirements need more clarity. We would also believe there should be a playing field level so that software functions that meet the definition of a medical device are regulated the same way regardless of whether they are deployed in a mobile application, an electronic health record (EHR), or traditional medical device.

Additional guidance on applying HIPAA, de-identification standards, and secondary use rules to AI training, validation, and model improvement would be beneficial. We encourage reducing uncertainty around permissible uses of clinical data for model improvement including the use of real-world data to support postmarket learning and iterative improvement. Furthermore, we recommend that postmarket monitoring be outcome-focused and risk-based rather than blanket requirements that could slow clinical adoption or disproportionately burden high-performing, low-risk technologies.

3. ***For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?***

Legal and implementation issues include liability allocation among developers, deployers, and clinicians, indemnification and contractual risk-shifting, and transparency expectations for AI-assisted decisions.

For AI tools used in clinical settings that are not regulated as medical devices, the biggest issue is uncertainty about responsibility. When these tools influence documentation, workflow, triage, or other aspects of care, it is not always clear who is accountable if something goes wrong or how existing privacy and compliance rules apply. That ambiguity can slow adoption, even when the technology is beneficial. HHS does not need to create new liability structures, but clearer guidance on how existing rules apply would reduce uncertainty and support responsible use.

An additional emerging issue involves the distinction between algorithm developers and model deployers. In some cases, companies provide foundational algorithms or configurable AI systems, while health systems or other end users perform the training and generate site-specific models. Current guidance often focuses on the evaluation and oversight of deployed models, leaving ambiguity regarding:

- The liability boundary between the underlying algorithm and the trained model
- The level of testing and validation expected at the algorithm stage versus the trained-model stage
- The allocation of responsibility for performance, bias mitigation, and post-deployment monitoring

This distributed development paradigm challenges traditional governance frameworks that assume a single entity controls design, training, and deployment.

HHS could play a valuable role by clarifying shared-responsibility expectations across the AI lifecycle and providing guidance on proportional testing and validation requirements at each stage. Such clarity would reduce contractual uncertainty, promote innovation, and ensure accountability without discouraging flexible AI development models.

4. ***For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?***

Any evaluation methods promoted by HHS should align with existing regulatory and quality systems and avoid duplicative third-party validation, redundant documentation, or unnecessary disclosure of confidential commercial information or sensitive health data.

5. ***How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?***

All medical devices, including AI-driven medical devices, are already highly regulated across the entire product lifecycle, including both the pre- and post-market timeframes. Devices are subject to robust post-market requirements that ensure their ongoing safety and effectiveness. When considering additional post-market requirements for emerging AI technologies, it is important that any new measures are tailored to specific device risks and characteristics, rather than applied as broad, blanket requirements to all AI-enabled devices. It is also important to recognize that AI-enabled devices are not uniform in function or risk. Governance frameworks, including hospital-based AI review committees, should apply

proportionate, risk-based oversight that reflects the device's intended use, level of autonomy, and degree of clinical oversight, rather than treating all AI technologies equivalently. Additionally, third-party certification should be avoided, as it will lead to redundant validation activities and documentation and may lead to disclosure of commercial confidential information and inappropriate release of private healthcare data.

6. *Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?*

AI has met or exceeded expectations as it relates to imaging and diagnostics, laboratory workflow optimization, and supply chain visibility.

AI has fallen short where tools were deployed without sufficient workflow integration, where performance was evaluated on retrospective datasets that did not reflect real-world variability, or where limited interoperability constrained scalability. Unrealistic expectations regarding full automation, rather than augmentation of clinical decision-making, have also contributed to underperformance in some settings.

The ability to more fully evaluate and scale AI tools across care settings is directly influenced by access to high-quality, interoperable data. Broader use of secure, standardized, and privacy-protective data sharing agreements would enable more robust validation across diverse populations and clinical environments. Such access would support stronger evidence generation, improve generalizability, and allow clearer demonstration of clinical and economic value.

7. *Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?*

Chief information security officers, chief technology officers, chief innovation officers, quality assurance, information technology, hospital information technology, procurement, revenue cycle management, engineering, and laboratory directors have the most influence on the adoption of AI for clinical care. Hurdles include supply chain, data sharing agreements, security, transparency, risk management, change management, validation, contract/legal review, reimbursement, and procurement.

8. *Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.*

Greater interoperability between laboratory systems, diagnostic platforms, and downstream clinical systems (including the imaging-to-EHR interoperability gap) would unlock substantial AI value. Standards like FHIR, DICOM and DICOM-SR can help, but they have not been adopted consistently. Technical and performance standards must be identified. Allowing cross-system integration will enable imaging-based AI tools to avoid being siloed within radiology workflows limiting downstream clinical adoption, multidisciplinary

decision-making, and development of value-based reimbursement models. Data cleanliness and privacy should be prioritized. Finally, benchmarking by key opinion leaders would be useful, as well as background benchmarking, though issues exist with the need to upskill to use AI to detect issues and deskill prevention of domain knowledge.

9. *What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?*

Patients and caregivers consistently express interest in AI tools that meaningfully improve access, timeliness, and quality of care. Desired outcomes include faster and more preventive diagnostics, better outcomes, reduced delays in testing and treatment, improved coordination and bedside decision support, and better prediction of system downtime or care disruptions that may affect access. AI that reduces administrative burden and enables clinicians to spend more time with patients is also viewed positively.

At the same time, patients and caregivers raise important concerns related to privacy, data security, bias, overreliance on automated systems, and the potential for malicious cyber activity. These concerns are appropriate and should continue to be addressed transparently.

Importantly, many of these risks are already actively managed within existing regulatory and quality frameworks. AI-enabled medical devices are subject to established cybersecurity requirements, including secure design controls, vulnerability management, and post-market monitoring to guard against malicious activity. Similarly, the risk of bias, while not unique to AI, is addressed through data representativeness assessments, performance testing across relevant populations, and risk mitigation strategies during development. For regulated products, these efforts are evaluated as part of premarket review and ongoing quality system oversight.

For non-device AI deployed in clinical environments, organizations similarly integrate privacy, security, and performance monitoring controls within existing governance and compliance programs. Continued transparency, lifecycle monitoring, and human oversight remain essential to maintaining public trust.

By strengthening interoperability, reinforcing cybersecurity standards, and promoting responsible development practices, AI can address patient priorities while operating within established safeguards designed to protect safety, equity, and confidentiality.

10. *Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?*

Priorities include trustworthy AI, bias mitigation across populations, human-AI collaboration models, real-world effectiveness and cost-benefit analysis, implementation science, evaluation frameworks, public private partnerships and literature gaps.

Several of these areas (particularly evaluation frameworks, robustness testing methods, and transparency standards) are well suited for development through international consensus standards bodies and global harmonization efforts. Leveraging international standards

promotes consistency, reduces duplicative regulatory burden, and supports global interoperability of AI-enabled healthcare technologies.

Accordingly, HHS research investments should complement, not replace, international standards development. The Department can play a valuable role by funding applied research, supporting public-private partnerships, and participating in global harmonization initiatives to ensure that U.S. regulatory and policy approaches remain aligned with internationally recognized best practices.

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We appreciate the opportunity to comment on this important topic. If we can provide any additional information, please contact me at mleahey@medicaldevices.org or (202) 354-7171.

Sincerely,

A handwritten signature in dark blue ink, reading "Mark B. Leahey". The signature is fluid and cursive, with a long horizontal stroke extending to the right.

Mark Leahey
President & CEO
Medical Device Manufacturers Association